

Linear Algebra Done Wrong (Sergei Treil) 1.1

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1. Let $\mathbf{x} = (1, 2, 3)^T$, $\mathbf{y} = (y_1, y_2, y_3)^T$, and $\mathbf{z} = (4, 2, 1)^T$. Compute $2\mathbf{x}$, $3\mathbf{y}$, $\mathbf{x} + 2\mathbf{y} - 3\mathbf{z}$

$$2\mathbf{x} = (2, 4, 6)^T$$

$$3\mathbf{y} = (3y_1, 3y_2, 3y_3)^T$$

$$\mathbf{x} + 2\mathbf{y} - 3\mathbf{z} = (1, 2, 3)^T + (2y_1, 2y_2, 2y_3)^T + (-12, -6, -3) = (2y_1 - 11, 2y_2 - 4, 2y_3)$$

2. Which of the following sets (with natural addition and multiplication by a scalar) are vector spaces. Justify your answer.

- (a) The set of all continuous functions on the interval $[0, 1]$.

Yes, this is a vector space: Commutativity:

$$f + g = g + f$$

Associativity:

$$(f + g) + h = f + (g + h)$$

Zero vector:

We take $f(x) = 0$ to be the zero vector. Then, for all g which are in the vector space, $f + g = g$

Additive inverse:

For every vector $f(x)$ we take $g(x) = -f(x)$ which gives $(f + g)(x) = 0$ for all x , $f + g$ is indeed the 0 vector.

Multiplicative Identity:

$$1f(x) = f(x)$$

Multiplicative Associativity:

$$(\alpha\beta)f = \alpha(\beta f)$$

Distributive Properties:

$$\alpha(f + g) = \alpha f + \alpha g$$

for all $f, g \in V$ and $\alpha \in R$.

$$(\alpha + \beta)f = \alpha f + \beta f$$

for all $\alpha, \beta \in R$ and $f \in V$.

Since we satisfy all the axioms, this is indeed a vector space.

- (b) The set of all non-negative functions on the interval $[0, 1]$.

This set is not closed under scalar multiplication: If f is a nonnegative function on the interval $[0, 1]$, then $g(x) = -f(x)$ is not a nonnegative function.

- (c) The set of polynomials of degree *exactly* n .
This is not a vector space, as there is no 0 vector. If $P(x)$ and $Q(x)$ are polynomials of degree n , then $P(x) + Q(x) \neq P(x)$ as the x^n term in $P(x)$ cannot be the same as the x^n term in $Q(x)$, as that would imply $Q(x)$ is a polynomial of degree at most $n - 1$, which cannot be the case.

- (d) The set of all symmetric $n \times n$ matrices, i.e. the set of matrices $A = \{a_{j,k}\}_{j,k=1}^n$ such that $A^T = A$

Similar to part b, if we are to check all axioms of a vector space on this set, we will find that they are all indeed satisfied. Another way of looking at this is the set of all symmetric $n \times n$ matrices is essentially the same as $R^{\frac{n(n+1)}{2}}$, since a symmetric matrix is defined by all elements including and below the main diagonal of the matrix.

3. True or False:

- (a) Every vector space contains a zero vector.
true, it's an axiom of vector spaces.
- (b) A vector space can have more than one zero vector.
false, we prove this in the following question.
- (c) An $m \times n$ matrix has m rows and n columns.
true, by definition.
- (d) If f and g are polynomials of degree n , then $f + g$ is also a polynomial of degree n .
false, let $n = 1$ and $f(x) = x + 1$ and $g(x) = -x$: $(f + g)(x) = 1$, which is not a polynomial of degree 1.
- (e) If f and g are polynomials of degree at most n , then $f + g$ is also a polynomial of degree at most n .
true, $f + g$ cannot possibly have a term x^k such that $k > n$, since that would imply either f or g would contain such a term, contrary to the fact that both are degree n . Thus, $f + g$ is at most degree n .

4. Prove that a zero vector $\mathbf{0}$ of a vector space is unique.

Let x and y be zero vectors in a vector space. If we consider the sum $x + y$, then $x + y = x$ must hold as y is a zero vector, and $x + y = y$ must also hold since x is a zero vector. If $x \neq y$, then both statements cannot be true at the same time, meaning the zero vector is unique.

5. What matrix is the zero vector of the space $M_{2 \times 3}$?

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

For any matrix of the form $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$, we have

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

Thus, $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is indeed the zero vector.

6. Prove that the additive inverse, defined in Axiom 4 of a vector space is unique.

For a vector $\mathbf{v}$, let $\mathbf{a}$ and $\mathbf{b}$ be additive inverses of $\mathbf{v}$.

We have:

$$\mathbf{v} + \mathbf{a} = \mathbf{0}$$

$$\mathbf{v} + \mathbf{b} = \mathbf{0}$$

If we add $\mathbf{a}$ to the second equation we get

$$\mathbf{v} + \mathbf{b} + \mathbf{a} = \mathbf{a}$$

Since vector addition is both commutative and associative, we get

$$(\mathbf{v} + \mathbf{a}) + \mathbf{b} = \mathbf{a}$$

Giving

$$\mathbf{b} = \mathbf{a}$$

7. Prove that $0\mathbf{v} = \mathbf{0}$ for any vector $\mathbf{v} \in V$.

By the distributive property, we know

$$1\mathbf{v} + 0\mathbf{v} = (1 + 0)\mathbf{v} = \mathbf{v}$$

since $\mathbf{v} + 0\mathbf{v} = \mathbf{v}$, it must be the case that $0\mathbf{v}$ is the zero vector, and because the zero vector is unique $0\mathbf{v} = \mathbf{0}$.

8. Prove that for any vector $\mathbf{v}$ its additive inverse $-\mathbf{v}$ is given by $(-1)\mathbf{v}$.

The additive inverse of a vector $\mathbf{v}$ is a vector $-\mathbf{v}$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$. Furthermore, this vector is unique.

By the distributive property, we have

$$1\mathbf{v} + (-1)\mathbf{v} = (1 + (-1))\mathbf{v} = 0\mathbf{v} = \mathbf{0}$$

From the previous question, we know $0\mathbf{v} = \mathbf{0}$.